TECHNICAL DOCUMENTATION

Real-Time Ferry Ticket Sales & Redemption Analytics for Toronto Island Park

Detailed guide and project requirements for the Real-Time Ferry Ticket Sales & Redemption Analytics for Toronto Island Park analysis.

Context:

Ferries operate year-round from Jack Layton Ferry Terminal to Centre Island, Hanlan's Point, and Ward's Island. Understanding near real-time passenger movement is critical for operational efficiency, staffing, and public safety.

Unified Mentor

Toronto Government Parks, Forestry & Recreation

Problem Statement

Despite having rich customer-level data, banks often lack:Although ferry ticket sales and redemptions are recorded every 15 minutes, there is no centralized analytics system to:

- • Identify peak passenger flow windows
- • Track net passenger movement
- • Support data-driven operational planning

Primary Objectives

- Analyze historical and near real-time ticket activity
- Identify peak and off-peak ferry demand periods
- Quantify passenger inflow and outflow trends

Secondary Objectives

- Support operational decision-making for ferry scheduling
- Improve passenger experience during peak hours

Dataset Description

COLUMN	DESCRIPTION
_id	Unique row identifier
Timestamp	Tickets sold in interva
Redemption Count	Tickets redeemed in interval

Analytical Methodology (Step-by-Step)

Data Ingestion

- Load CSV data (2015–2025)
- Convert timestamps to datetime
- Sort chronologically

Data Cleaning

- Missing timestamp handling
- Outlier detection in counts
- Data consistency checks

Data Cleaning

- Missing timestamp handling
- Outlier detection in counts
- Data consistency checks

Feature Engineering

- Hour of day
- Day of week
- Month / season
- Weekend vs weekday

:Exploratory Data Analysis (EDA)

- Hourly and daily demand trends
- Seasonal comparison
- Sales vs redemption distribution
- Rolling averages (1-hour, 4-hour)

Key Performance Indicators (KPIs)

- Tickets Sold per Hour
- Tickets Redeemed per Hour
- Net Passenger Movement (Sales - Redemptions)
- Peak Demand Windows
- Off-Season Utilization Index

Streamlit Web Application Requirements

Dashboard Modules

- Real-time KPI cards
- Interactive time-series plots
- Date & time filters
- Peak vs off-peak comparison

User Roles

- Operations team
- Policy planners
- Management stakeholders

Streamlit Web Application Requirements

Dashboard Modules

- Real-time KPI cards
- Interactive time-series plots
- Date & time filters
- Peak vs off-peak comparison

User Roles

- Operations team
- Policy planners
- Management stakeholders

Deliverables and Submission

- Research paper (EDA, insights, recommendations)
- Streamlit dashboard (live analytics)
- Executive summary for government stakeholders

Expected Impact

- Improved ferry scheduling
- Reduced congestion at terminals
- Data-driven operational planning